② Multiple Regression.

area	bedroom	age	price
2600	3	20	550000
3000	4	15	565000
3200		18	610000
3600	3	30	595000
4000	5	8	760000

Find out the price of a home that has,
3000 sqr ft area, 3 bedrooms, 40 year old

dependent

independent

price = m_1 * area + m_2 * bedrooms + m * age + b

coefficients

$$\boxed{y = m_1x_1 + m_2x_2 + m_3x_3 + b}$$

```
import pandas as pd
import numpy as np
from sklearn import linear_model

df = pd.read_csv("homeprices.csv")
df
import math
median_bedrooms = math.floor(df.bedrooms.median())
median_bedrooms
df.bedrooms.fillna(median_bedrooms)
```

$\Big\}$ data preprocessing

```
reg = linear_model.LinearRegression()
reg.fit(df[['area', 'bedrooms', 'age']], df.price)
reg.coef_
reg.intercept_
reg.predict([[3000, 3, 40]])
```